Project README

Geospatial Identification of Rooftop Solar Energy Potential Using Building Footprints and Solar Radiation Data

Case Study: San Diego, California

1. Project Overview

This project aims to identify rooftops with strong potential for solar energy installation in San Diego, California using spatial data analysis. Rooftop solar systems are an effective way to generate renewable energy in urban areas without requiring additional land resources. However, not all buildings are equally suitable for solar installation due to differences in rooftop size, spatial distribution, and solar exposure.

The objective of this project is to analyze building footprints and solar radiation data to estimate rooftop solar potential across the city. By integrating open geospatial datasets and performing spatial analysis in Python, the study will identify buildings that are most suitable for rooftop solar systems.

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2. Study Area

City: San Diego, California, United States

San Diego is selected because it has high solar potential and publicly available geospatial datasets that support spatial data science analysis.

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3. Datasets Used

3.1 Building Footprints Dataset

Source: Microsoft US Building Footprints

Download Link:
https://github.com/microsoft/USBuildingFootprints

Dataset Used:
California Building Footprints

Format:
GeoJSON (.geojson)

Description:
This dataset contains building polygons representing rooftop outlines across the United States. Each polygon represents the shape of an individual building. These polygons are used to identify rooftop structures and calculate rooftop area.

Purpose in Project:
• Identify building rooftops
• Calculate rooftop area
• Estimate rooftop solar installation potential

3.2 City Boundary Dataset

Source: U.S. Census TIGER/Line Shapefiles

Download Link:
https://www2.census.gov/geo/tiger/TIGER2023/PLACE/tl_2023_06_place.zip

Format:
Shapefile (.shp)

Description:
This dataset contains geographic boundaries for all cities in California. The dataset will be filtered to extract the boundary polygon for San Diego.

Purpose in Project:
• Define the study area
• Clip buildings within the San Diego boundary
• Limit analysis to the selected city

3.3 Solar Radiation Dataset

Source: Global Solar Atlas / Solargis

Dataset:
Long-Term Average Global Horizontal Irradiation (GHI)

Format:
GeoTIFF (.tif)

Description:
This raster dataset contains solar radiation values measured as Global Horizontal Irradiation (GHI). Each raster pixel represents the annual solar radiation received in that location.

Unit:
kWh per square meter per year ($kWh/m^2/year$)

Purpose in Project:
• Measure solar exposure across the city
• Estimate rooftop solar energy potential
• Combine solar radiation values with rooftop polygons

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4. Software and Tools

Programming Language:
Python

Main Libraries:
• GeoPandas – spatial vector data processing
• Rasterio – raster dataset handling
• Shapely – geometry operations
• NumPy – numerical analysis
• Pandas – tabular data analysis
• Matplotlib – visualization

Optional Tools:
• QGIS for data inspection and visualization

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5. Project Methodology

Step 1 – Data Preparation

All datasets will be loaded into Python and inspected. Coordinate reference systems will be standardized to ensure compatibility between datasets.

Step 2 – Study Area Extraction

The California city boundaries dataset will be filtered to extract the polygon corresponding to San Diego. This polygon will define the spatial boundary for the analysis.

Step 3 – Building Footprint Processing

The Microsoft building footprint dataset will be clipped using the San Diego boundary polygon. This step ensures that only buildings located within the city are included in the analysis.

For each building polygon, the rooftop area will be calculated using geometric operations.

Step 4 – Solar Radiation Processing

The solar radiation raster dataset will be loaded and clipped to the San Diego boundary. This step reduces the dataset to only the relevant geographic area.

Each raster pixel contains the solar radiation value representing the potential solar energy available at that location.

Step 5 – Spatial Overlay Analysis

Building polygons will be spatially overlaid with the solar radiation raster. Using zonal statistics or spatial sampling techniques, the solar radiation value corresponding to each building rooftop will be extracted.

Step 6 – Rooftop Solar Potential Estimation

A simple solar suitability indicator will be calculated using rooftop area and solar radiation values. Larger rooftops with higher solar exposure will have higher solar energy potential.

Buildings may be classified into categories such as:

• High solar potential
• Moderate solar potential
• Low solar potential

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6. Expected Outcomes

The project will produce:

• A spatial dataset containing rooftop solar potential estimates
• Maps showing the distribution of rooftop solar suitability
• Identification of areas with high solar installation potential

These results will demonstrate how spatial data science techniques can support renewable energy planning in urban environments.

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7. Project Significance

This study demonstrates how open geospatial datasets and Python-based spatial analysis can be used to evaluate renewable energy potential in cities. The results can help highlight areas where rooftop solar adoption could be most effective and illustrate the usefulness of geospatial data for sustainable urban planning.